Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	15 March 2026
Team ID	PNT2026TMIDxxxxxx
Project Name	AI-Enabled Student Admissions & Support Management System Using Salesforce Agentforce
Maximum Marks	4 Marks

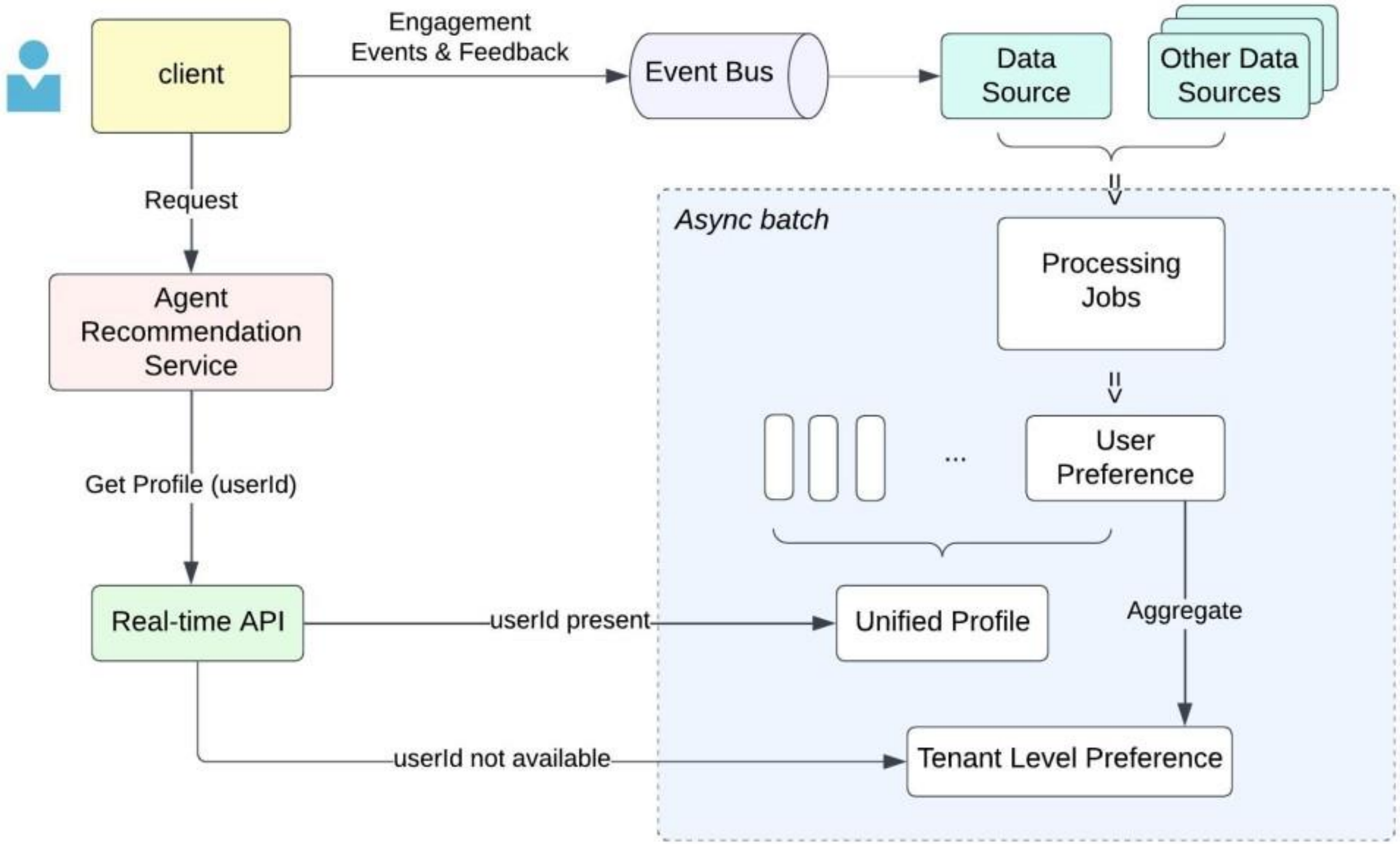
Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Agent Recommendation High Level Architecture

Reference: https://engineering.salesforce.com/solving-llm-challenges-agentforces-ai-approach-to-reliable-agent-recommendations/

Agent Recommendation High Level Architecture



S.No	Component	Description	Technology
1	User Interface	How students, staff, and administrators interact with the system (web portal, mobile app, chatbot).	Salesforce Lightning Web Components (LWC), HTML, CSS, JavaScript, Mobile App (Salesforce Mobile SDK)
2.	Application Logic-1	Core business logic for admissions workflows (applications, approvals, enrollment).	Salesforce Apex, Flow Builder, Process Builder
3.	Application Logic-2	AI-driven student support (query resolution, FAQs, personalized guidance).	Salesforce Agentforce AI, Einstein Bots, NLP models
4.	Application Logic-3	Predictive analytics for admissions trends, student success, and support needs.	Salesforce Einstein Analytics, Machine Learning models
5.	Database	Storage of student records, applications, and support tickets.	Salesforce Data Cloud, Relational DB (PostgreSQL/MySQL if external integration)
6.	Cloud Database	Cloud-based scalable storage for admissions and support data.	Salesforce Data Cloud, AWS RDS (if hybrid integration)
7.	File Storage	Storage of documents (transcripts, certificates, ID proofs).	Salesforce Files, AWS S3, Google Drive integration
8.	External API-1	Integration with external services (education boards, payment gateways, verification).	REST APIs, SOAP APIs, Government Education APIs, Stripe/PayPal for payments

9.	External API-2	AI services for advanced NLP and speech-to-text for student queries.	OpenAI API, Google Cloud Speech-to-Text, IBM Watson STT
10.	Machine Learning Model	Predictive models for admissions trends, student success, and personalized support.	Predictive ML (Einstein Analytics, TensorFlow, Scikit-learn)
11.	Infrastructure (Server / Cloud)	Core platform and cloud environment to run applications, AI services, and integrations.	Salesforce Platform, AWS Cloud, Docker/Kubernetes

S.No	**Characteristics**	**Description**	**Technology**
1.	Open-Source Frameworks	Frameworks used for UI and ML integration	TensorFlow, Scikit-learn, React (if external UI)
2.	Security Implementations	Authentication, encryption, and access controls for student data	Salesforce Shield, IAM, OAuth 2.0, TLS/SSL, OWASP standards
3.	Scalable Architecture	Supports growing admissions and support load with modular design	Salesforce multi-tenant cloud, Microservices, API-first design

4.	Availability	Ensures system uptime and reliability with redundancy and load balancing	Salesforce Cloud, AWS Load Balancer, Distributed Servers

S.No	Characteristics	Description	Technology
5.	Performance	Optimized for high request volumes, caching, and fast response times	Salesforce CDN, Redis Cache, Query Optimization

References:

Salesforce Architects – Welcome Architects

Improving Admissions with Salesforce Education Cloud (salesforce.com in Bing)

AWS Architecture Center (aws.amazon.com in Bing)

OWASP Foundation